

EXAM 7 – SPRING 2012

3. (2.5 points)

Given the following information as of December 31, 2011:

<u>Cumulative Incurred Losses (\$000)</u>		
Accident Year	As of 24 Months	As of 36 Months
2004	2,000	2,600
2005	4,000	6,000
2006	6,500	11,700
2007	6,000	10,000
2008	3,600	6,600
2009	7,600	11,000
2010	5,000	

a. (2 points)

Using a volume-weighted average to calculate the overall age-to-age factor, create a plot of weighted residuals following Mack's methodology.

b. (0.5 point)

Based on the residual plot, assess whether the variance assumption underlying the chain ladder method has been met.

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Question 3 Sample Answer

Solution 1

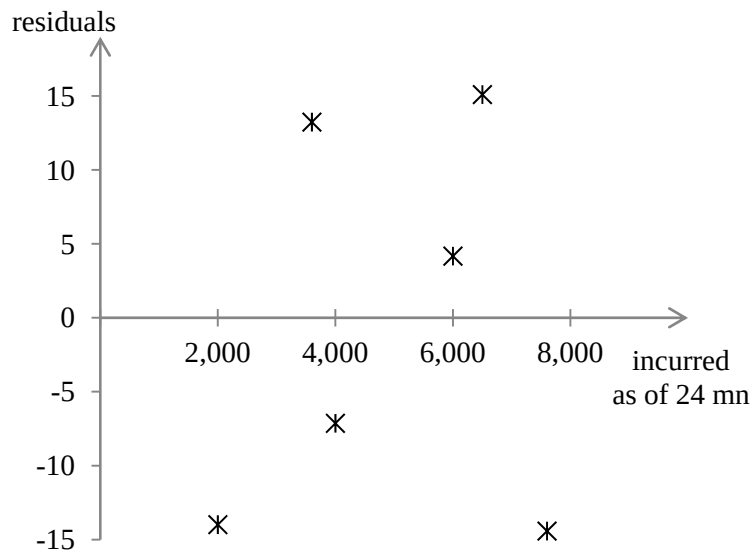
a)

$$\text{Mack residuals for volume weighted average} = \frac{C_{i,k+1} - C_{ik} f_k}{\sqrt{C_{ik}}}$$

$$f = \frac{2600 + 6000 + \dots + 11000}{2000 + 4000 + \dots + 7600} = \frac{\sum \text{cuml losses as of 36 months}}{\sum \text{cuml losses as of 24 months}}$$

$$= \frac{47,900}{29,700} = 1.613$$

AY	residuals		Inurred as of 24 mths
04	-14	← $\frac{2600 - 2000 \times 1.613}{\sqrt{2000}}$	2000
05	-7		4000
06	15		6500
07	4		6000
08	13		3600
09	-14		7600



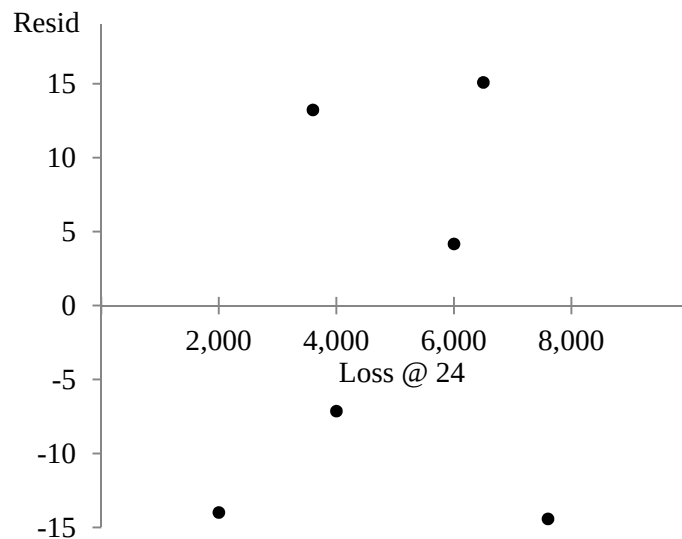
An underlying assumption of the chain ladder method is linearity of the development. We expect to see residuals scattered evenly about the y-axis. These residuals do appear randomly scattered about the x-axis $\therefore$ the variance assumption is met.

Solution 2

a)

$$\begin{aligned} \text{Age-Age} &= \frac{(2600 + \dots + 11,000)}{(2000 + \dots + 7600)} \\ &= \frac{47900}{29700} = 1.613 \end{aligned}$$

Yr	(a) 24 loss $\times$ 1.613	$\frac{\text{Loss@36} - (a)}{\sqrt{\text{Loss@24}}}$	
4	3,226	-14	$= (2000 - 3226) / \sqrt{2000}$
5	6,452	-7.15	
6	10,485	15.07	
7	9,678	4.16	
8	5,807	13.22	
9	12,259	-14.44	



b) The points are fairly randomly scattered around 0. They don't seem to be increasing or decreasing. So it appears the variance assumption has been met.

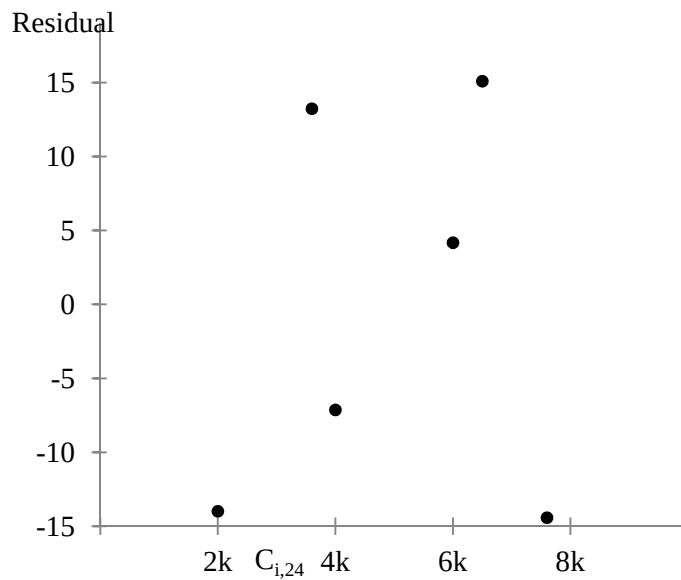
Solution 3

a)

$$\begin{aligned} 24 \text{ to } 36 \text{ LDF} &= \frac{\sum \text{losses @ 36 months years '04 to '09}}{\sum \text{losses @ 24 for years '04 to '09}} \\ &= 1.613 \end{aligned}$$

$$\hat{C}_{ij} = 1.613(C_{ij-1}) \quad \text{residual} = \frac{C_{i36} - \hat{C}_{i36}}{\sqrt{C_{i24}}}$$

$C_{i,36}$	$\hat{C}_{i36}$	Residual
2600	3226	-14.00
6000	6452	-7.15
11700	10484.5	15.08
10000	9678	4.16
6600	5806.8	13.22
11000	12258.8	-14.94



- b) The residuals should be random around the zero, with no clear patterns as $C_{i,24}$ increases. This appears to be the case. So we can assume the variance assumption is met.

Examiner Comment

Most candidates knew how to calculate the weighted LDF. There were a number of variations in the form of the residual calculation. Partial credit was given if the residual numerator had Actual – Expected. Many candidates did not use the square root of the 24 month actuals in the denominator. The graph was accepted if the plot was of residuals vs. loss @ 24 months, even if residuals were calculated incorrectly. For the assessment of the variance assumption, candidates frequently lost at least partial credit for not fully assessing and explaining the reasoning behind their answer. Candidates are encouraged to state what the assumption is and how the graph did or did not meet the criteria.